



AI POWERED RECONCILIATION AGENT
PRESENTED TO
AWS DISTRIBUTION

TABLE OF CONTENTS

EXECUTIVE SUMMARY	4
SECTION 01 – COMPANY PROFILE OVERVIEW.....	6
BACKGROUND	6
VISION 6	
OUR LEGACY.....	6
OUR GLOBAL PRESENCE	6
KEY DIFFERENTIATORS.....	7
OUR SERVICES AND TECHNOLOGY PARTNERS	8
TOP AWARDS AND RECOGNITIONS.....	11
SECTION 02 – ABACUS GCP/APIGEE REFERENCES	13
SECTION 03 – SCOPE OF WORK	18
3.1 SOLUTION OVERVIEW.....	18
3.2 HIGH-LEVEL ARCHITECTURE	18
3.3 SOLUTION ARCHITECTURE.....	18
3.4 GOOGLE CLOUD LANDING ZONE.....	20
3.5 RECONCILIATION LOGIC.....	21
3.6 USER JOURNEY	22
3.7 SECURITY AND ACCESS MANAGEMENT	22
3.8 DELIVERABLES	23
3.9 OUT OF SCOPE	23
SECTION 04 – IMPLEMENTATION PLAN AND TIMELINE	24
4.1 DELIVERY APPROACH	24
4.2 IMPLEMENTATION PHASES.....	24
4.3 MILESTONES.....	ERROR! BOOKMARK NOT DEFINED.
4.4 GOVERNANCE	25
4.5 CRITICAL SUCCESS FACTORS	25
SECTION 05 – GENERAL ASSUMPTIONS	25

EXECUTIVE SUMMARY

EXECUTIVE SUMMARY

AWS Distribution currently reconciles its Statement of Account against customer ledgers manually. Finance staff export both sides, align them in spreadsheets, and work line by line to identify matches, deductions, timing differences and disputes. The process is slow, difficult to audit, and the reasoning behind each decision is retained only by the individual who performed it.

Abacus Cambridge Partners proposes an AI Reconciliation Agent that automates this work end to end. Users log into a secure web portal, upload the AWS Distribution Statement of Account and the corresponding Customer Ledger as Excel files, and execute the match engine. The solution cleanses and normalises both datasets, applies a deterministic rule-based matching engine, categorises every unmatched item into a defined exception type, and then applies a Gemini-powered AI agent that explains each exception in plain English and recommends the best next action. Users review the exception queue, approve or adjust, and export a finalised reconciliation report.

The solution is delivered on Google Cloud Platform. Abacus will provision a minimal, purpose-built landing zone in the Doha region (me-central1) — the nearest Google Cloud region to the United Arab Emirates. The design is deliberately lean: only the services strictly required to run the solution are provisioned, with no unnecessary networking, analytics or security tiers that would inflate the monthly run rate.

The engagement is delivered entirely off-site over a six-week period. There is no integration with FastERP or any other source system, and no integration with an external identity provider. This keeps the delivery footprint small, the timeline short and the commercial envelope predictable, while giving AWS Distribution a production reconciliation capability that can be extended as requirements mature.

Key outcomes

- Reconciliation cycle time reduced from days of manual spreadsheet work to a single automated run.
- Every match and every exception recorded against a defined reason code, creating a defensible and repeatable audit trail.
- Plain-English AI explanations and recommended actions for each exception, reducing the level of expertise required to resolve them.
- A production-grade Google Cloud foundation owned by AWS Distribution and sized to actual need.

A futuristic cityscape at night, featuring a central skyscraper with a glowing blue digital overlay. The overlay displays various data points, charts, and icons, suggesting a high-tech or corporate theme. The city is reflected in a wet surface in the foreground.

COMPANY PROFILE OVERVIEW

SECTION 01 – COMPANY PROFILE OVERVIEW

BACKGROUND

Abacus Cambridge Partners – Middle East Limited (Abacus) is a technology consulting company that works with its customers to accelerate business transformation by converting ideas into innovative, secure, and scalable business solutions.

Headquartered in the United Kingdom, Abacus Cambridge Partners was established in 2017, as an exclusive member firm of Abacus Consulting Technology Ltd., formed in 1987, as a spin-off from PwC Consulting. Our DNA carries the rich and diverse technology consulting practices and experience of Cooper & Lybrand and PwC. With 11 practice areas, a team of over 4000, and a global footprint in 4 continents, through our regional offices and exclusive member firms:

- Abacus Cambridge Partners - UK
- Abacus Cambridge Partners Inc. – U.S.A
- Abacus Consulting Technology Ltd. – Pakistan
- Abacus Cambridge Partners Saudia LLC – K.S.A.

We have delivered the business transformation promise for more than 1500 global clients across 40 industries in 30 countries. We bring a unique blend of conglomerate diversity, start-up agility, and enterprise-grade solution delivery.

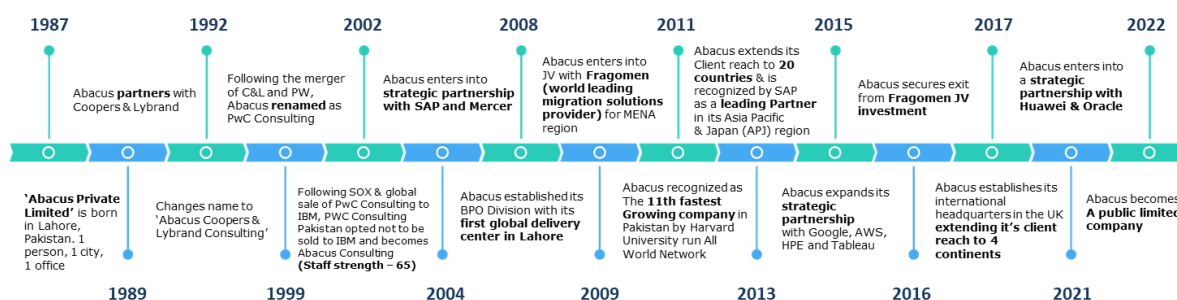
Today, we are a proud global technology solutions and outsourcing company, standing at the forefront of digital innovation and disruption.

Over our 35+ years journey we have evolved through a focus on building deep partnerships with global leaders and sustained investment in widening our geographical reach.

VISION

Be the transformative partner of choice globally for clients determined to succeed in a rapidly changing world, so that together we create the future of business.

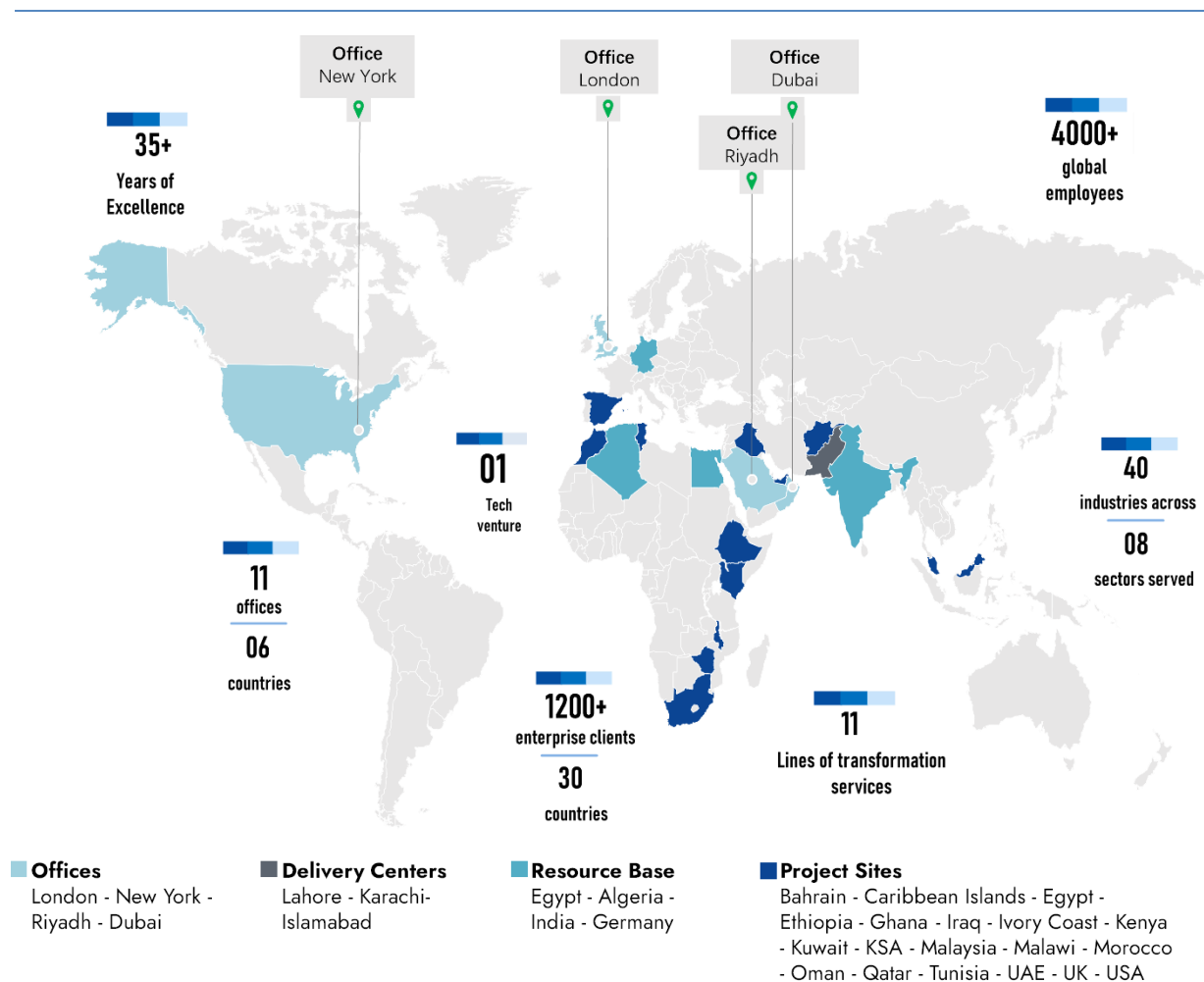
OUR LEGACY



OUR GLOBAL PRESENCE

Abacus is founded on one simple belief, that change is the only constant in our world and that as enterprises and individuals, we must all respond to change with vision, courage and integrity.

We at Abacus are committed to adding value to our clients, driven by a proven track record of building sustainable relationships across the globe. For us, fulfillment is seeing our clients transform into truly great organizations.

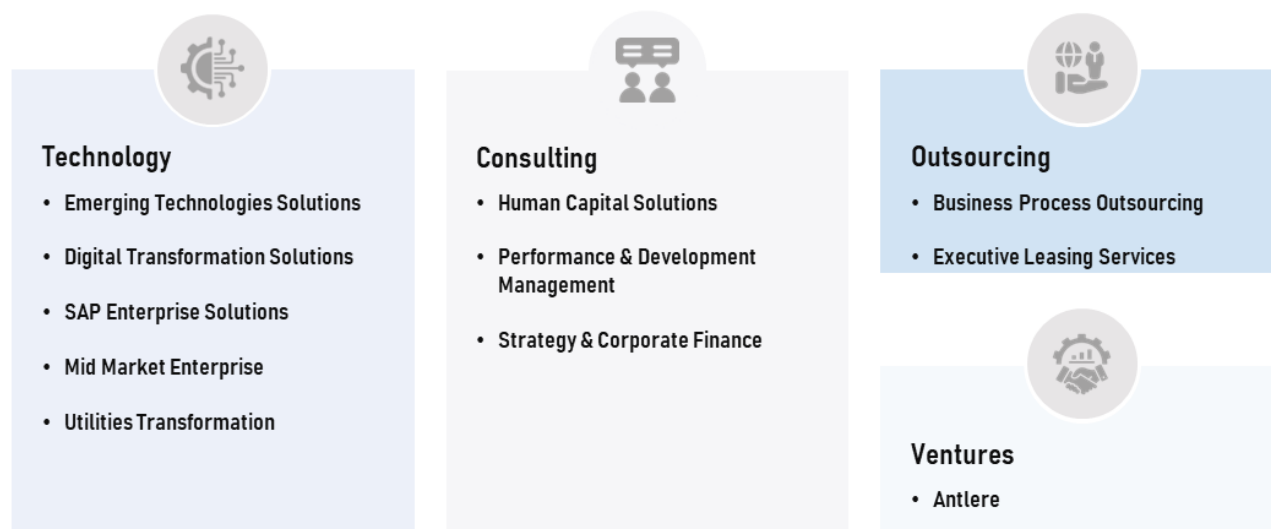


KEY DIFFERENTIATORS

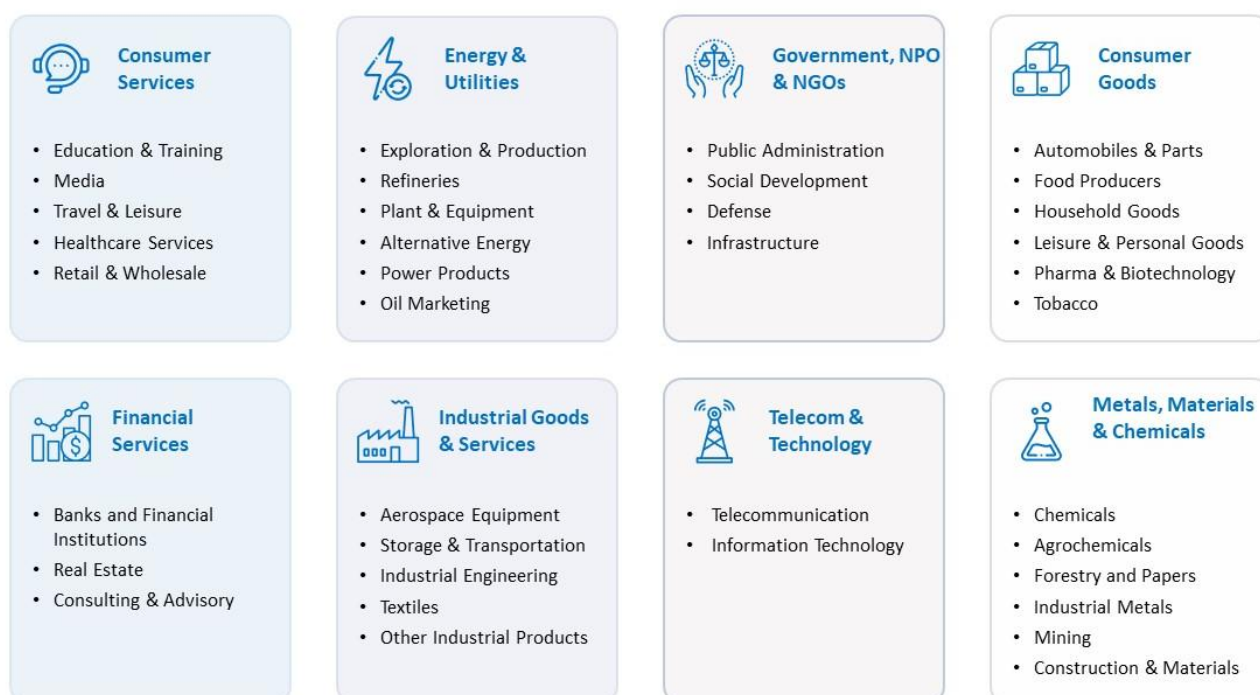
- Risk Management and International Certifications such as:
- ISO/IEC 27001:2022
- Technology Driven Solutions
- Diversified Solution Areas
- World Class Management Advisory Services
- Cross Industry and sector specialization
- Sustainable Client Relationships
- Offshore Delivery Capability
- Project Management Capability
- Corporate Social Responsibility

OUR SERVICES AND TECHNOLOGY PARTNERS

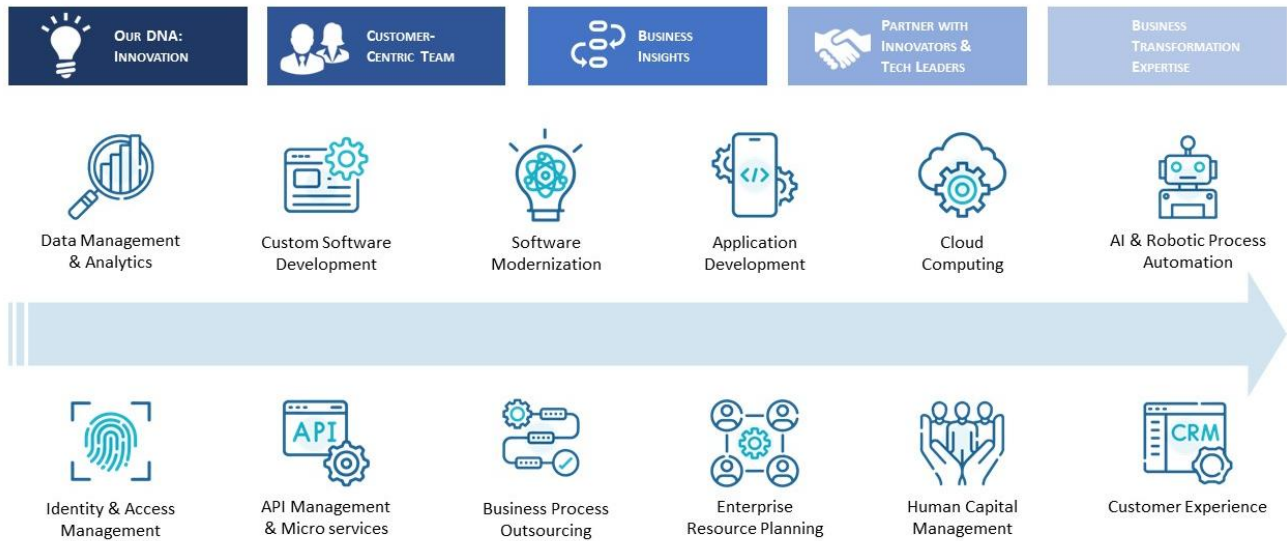
SERVICE PRACTICE AREAS



INDUSTRIES WE SERVE



OFFERED SOLUTIONS



OUR TECHNOLOGY PARTNERS

Strategic alliances lie as a cornerstone of our strategy. Through aligning with leading solution providers, we gain access to world-class business knowledge and technology which fuels our very ability to be a transformation partner for our clients.

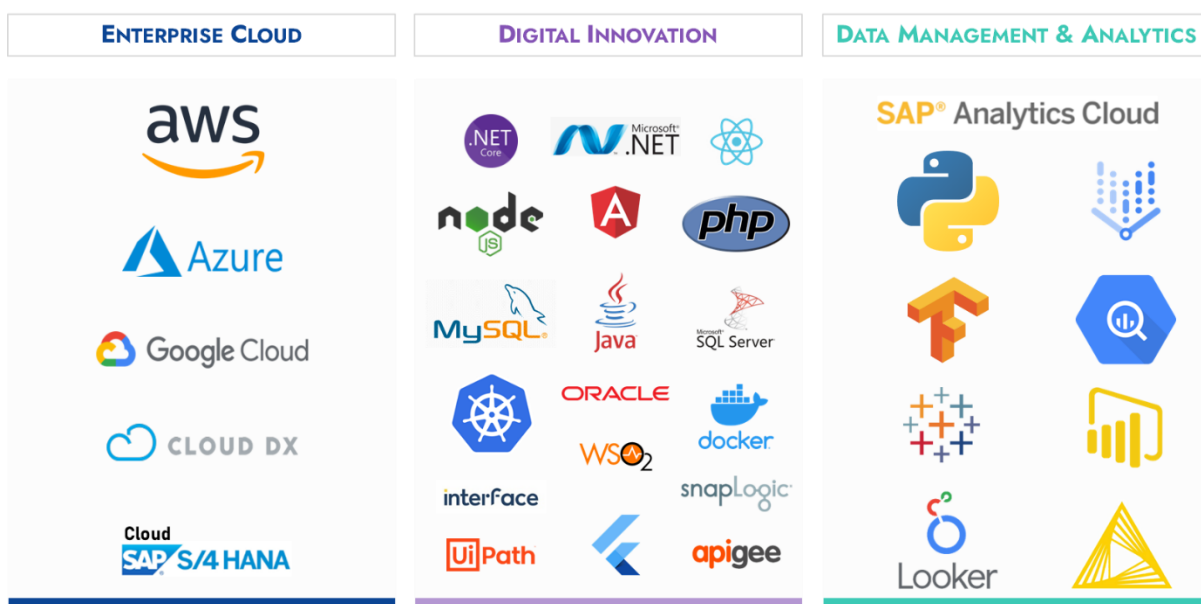


CONSULTING SOLUTION PARTNERS

Our core believes system revolves around collaborations for diversified and collective wisdom to serve the best interest of our clients. We therefore endeavor to form partnerships in delivering our services to ensure that our clients are served with bespoke solutions to address the business challenges of tomorrow.



OUR TECHNOLOGY STACK



TOP AWARDS AND RECOGNITIONS



SAP Top Partner of the Year MENA
2018 - 23



Google Cloud Premier
Partner 2022



Google Application Development
Specialization 2022



Best Business One SAP Partner
2023



Google Apigee Emerging Partner of
Year in EMEA 2020



LCCI President's IT
Award 2021- 2023



MEA North Partner IP of the Year
Award 2022



Highest SAP Business One HANA
Deals in MENA 2017



SAP Innovation Partner of the Year
MENA 2016



Largest SMB Partner Award MEA
North 2022



SAP Business One Best New Names
MENA 2017



Global Diversity, Equity & Inclusive
Benchmarks
Awards - 5 Times Winner



Best in ICT Services in Pakistan
2018



3rd Best SAP Business One Partner of
the Year EMEA 2017



ABACUS FRONT-END DEVELOPMENT REFERENCES

SECTION 02 – ABACUS GCP/APIGEE REFERENCES

Sr.No.	CLIENT'S NAME	LOCATION	PROJECT TITLE	DESCRIPTION
•	Central Agency for Information Technology (CAIT)  Kuwait Central Agency for information technology	Kuwait	Integration Platform Setup	The Central Agency for Information Technology (CAIT) is a government entity responsible for driving digital transformation, IT governance, and national technology initiatives. It oversees IT infrastructure, cybersecurity, and e-government services, ensuring seamless integration across public sector systems. CAIT also establishes technology policies, standards, and frameworks to enhance efficiency, security, and interoperability in government digital services. Project Details are confidential; the high level scope includes the following. Integration & Automation - Used Google Application Integration for ERP connectivity. Security & Compliance - Enforced IAM roles, LS encryption, and logging policies. Monitoring & Alerts - Integrated Cloud Logging & Monitoring for real-time insights. Scalability & Performance - Enabled auto-scaling, caching, and traffic optimization.
•	Financial Services Organization	Caribbean	Digital Transformation with Google Cloud	A diversified financial services organization based in the Caribbean. Its strategic business units (SBUs) span across banking, pensions, Wealth management. Money transfer. Property services. And philanthropy (Foundation). The Group is undergoing an ambitious transformation to become a digital-first, customer-centric market leader across the region.
•	Safaricom 	Ethiopia	Apigee Implementation	Safaricom is a leading communications company in Ethiopia, Abacus has been recently awarded the contract for supply and implementation of Apigee API management system. The details of the project are confidential at this stage.
•	Safaricom 	Kenya	Apigee Implementation and Monetization	Safaricom is a leading communications company in Kenya with the widest and strongest coverage. The home of the famous Mobile Money service- M-PESA and a leading innovator in mobile service. Safaricom selected Apigee to help them accelerate digital innovation. Abacus Cambridge Partners is part of this strategic Initiative and has completed the end to end implementation or the Apigee solution successfully.
•	Majid ul Futtaim 	UAE	Apigee Implementation	Majid Al Futtaim, is a leading retail company based in Dubai, UAE. Majid Al Futtaim operates in 15 countries across the Middle East, Africa and Asia through a diversified brand portfolio of shopping malls, communities, Grocery and leisure establishment. The group is also engaged in entertainment, healthcare, and food and beverage activities.

Majid ul Futtaim 	UAE	Apigee Governance and Security	Majid Al Futtaim, is a leading retail company based in Dubai, UAE. Majid Al Futtaim operates in 15 countries across the Middle East, Africa and Asia through a diversified brand portfolio of shopping malls, communities, Grocery and leisure establishment. The group is also engaged in entertainment, healthcare, and food and beverage activities.
Arab Banking Corporation (B.S.C.) 	Bahrain	Apigee Implementation	Bank ABC (incorporated as Arab Banking Corporation B.S.C) is an international bank headquartered in Manama, Kingdom of Bahrain. Bank ABC is a leading provider of Trade Finance, Treasury, Project & Structured Finance, Syndications, Corporate & Institutional Banking as well as Islamic Banking services. Bank ABC's network spreads across 17 countries in the Middle East and North Africa (MENA), Europe, the Americas and Asia.
Arab National Bank 	KSA	Apigee Implementation and Monetization	The Arab National Bank (ANB) is a major bank based in Riyadh, Saudi Arabia and listed on the Saudi Stock Exchange. It is among the top ten largest banks in the Middle East and has received an "A" rank from Standard and Poor's. It has 156 branches in Saudi Arabia.
EmaraTech 	Iraq	Apigee Implementation and Monetization	Details of the project are confidential as this stage as the project is currently under implementation, however the high level scope of work includes Installation of Apigee on Development and Production environment (with Monetization Basic Installation and Configurations)
Emirates Integrated Telecommunications Company (Du) 	UAE	Apigee Implementation	Du Telecom is the leading telecom company in the UAE with more than 6.5 million mobile customers (almost 50% market share), 555,000 fixed line subscribers, and 180,000 home service subscribers.

•	EmaraTech 	UAE	Apigee Implementation	Emaratech, part of the profile for the Investment Corporation of Dubai, is the leading online technology solutions and consulting company in the Ardo world.
•	Emirates Airline 	UAE	Apigee Developer Portal	Emirates, the world's fourth-largest airline by scheduled revenue passenger-kilometers flown and number of international passengers carried was founded in 1985 by the royal family of Dubai.
•	JS Bank 	Pakistan	Apigee Implementation	JS Bank selected Abacus Cambridge Partners to implement Apigee API Management Platform. Through this initiative, JS Bank is playing a vital role in the transformation of digital banking landscape in the country and help in creating unprecedented opportunities for developers, customers and partners.
•	National Bank of Pakistan/ Karandaaz 	Pakistan	Apigee Implementation	National Bank of Pakistan (NBP) is a major Pakistani commercial bank The bank provides both commercial and public sector banking services. It is a lead player in the debt equity market, corporate investment banking, retail and consumer banking, agricultural financing, treasury services. As of March 2017 it has 1,450 branches across Pakistan with assets of approximately USD 17.2 billion. The Bank also operates a network of 21 international branches, international Subsidiaries and 3 international Representative Offices. The Bank also has presence in the UK through United National Bank Ltd (UNBL), a banking institution incorporated in the UK.
•	Telenor 	Pakistan	Apigee Implementation	Telenor engaged Abacus Cambridge Partners as their local Apigee partner to undertake the responsibilities of the System Integrator for the API gateway project. Telenor Pakistan plans to innovate and digitally accelerate by using an API gateway, which will allow for new revenue streams, faster Innovation and more Innovative products for Telenor's existing and prospective customers.
•	Mobile Telecom Company Zain 	KSA	Apigee Implementation	ZAIN is the pioneer of mobile telecommunications in the Middle East. ZAIN is leading mobile voice and data services operator with a commercial footprint in 8 Middle Eastern and Africa countries with a workforce of over 6,000 providing a comprehensive range of mobile voice and data services to over 50 million active individual and business customers as of March 31, 2019.

•	Commercial Bank	Jamaica	Core Modernization & Digital Banking with GCP & Abacus	Abacus is working with one of the biggest commercial banks in Jamaica to modernize its core architecture, cloud infrastructure, and digital engagement through a multi-phase roadmap. The engagement includes enterprise architecture redesign, GCP migration, data/AI modernization, and development of a secure, scalable API ecosystem-paving the way for a future-ready, customer-centric banking model. Transformation Goals: - Enterprise Architecture Overhaul - Cloud Infrastructure Modernization - API Management via Apigee - AI & Data Modernization - Digital Identity & IAM
---	------------------------	---------	--	---

SCOPE OF WORK



SECTION 03 – SCOPE OF WORK

3.1 SOLUTION OVERVIEW

The AI Reconciliation Agent is a web-based application that reconciles the AWS Distribution Statement of Account against a Customer Ledger for a defined reconciliation period. Both files are supplied by AWS Distribution users as Excel workbooks, downloaded manually from the systems in which they already reside and uploaded through the portal.

The solution combines two complementary layers:

- A **deterministic rule-based reconciliation engine** that performs the matching itself, applying a defined and repeatable set of rules so that results are reproducible and auditable.
- An **AI reconciliation agent** built on Google Gemini that operates on the output of that engine, producing a plain-English explanation for each exception and a recommended next action.

This separation is deliberate. Matching decisions remain deterministic and explainable by rule; the AI layer provides interpretation and recommendation without altering the underlying arithmetic.

3.2 HIGH-LEVEL ARCHITECTURE

The solution is deployed as a containerised application on Google Cloud Run, with a separately hosted front end, a managed PostgreSQL database and Vertex AI providing model inference. All components reside within a single Google Cloud region.

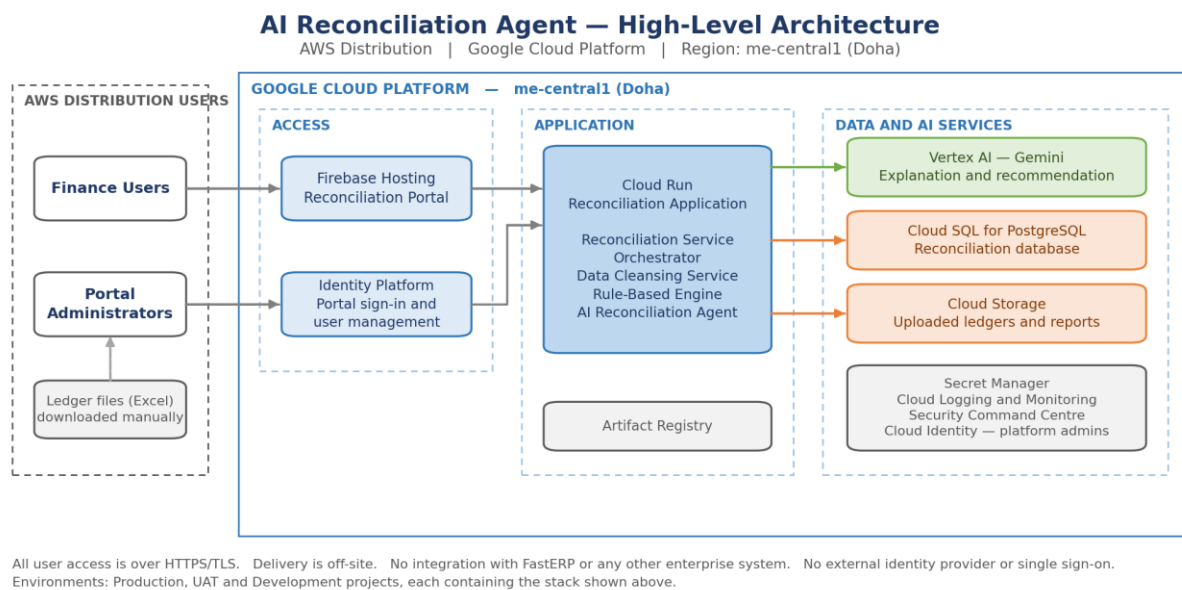


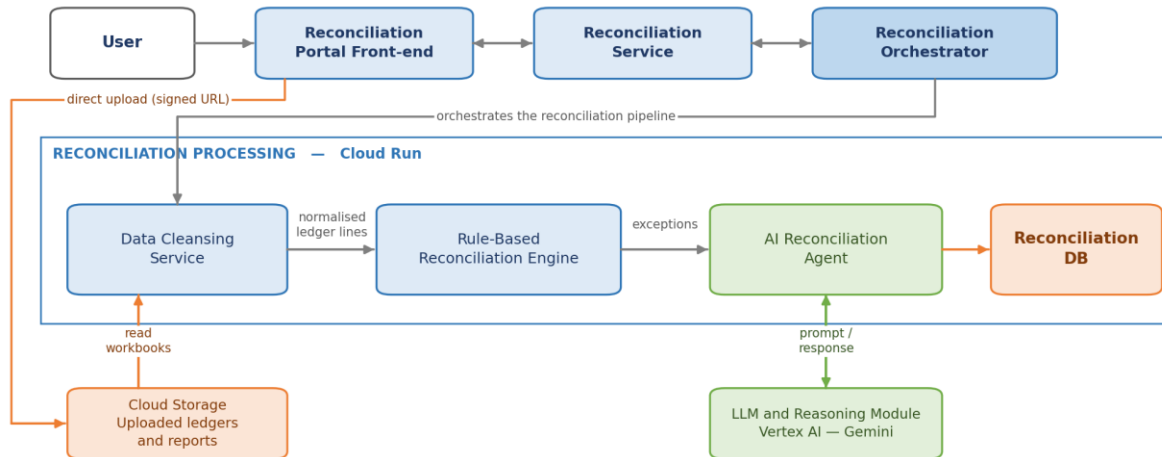
Figure 1 — AI Reconciliation Agent: high-level architecture on Google Cloud Platform

3.3 SOLUTION ARCHITECTURE

The reconciliation capability itself is composed of the components below. The Reconciliation Service exposes the application to the portal and owns persistence; the Orchestrator sequences the processing pipeline across the Data Cleansing Service, the Rule-Based Reconciliation Engine and the AI Reconciliation Agent.

AI Reconciliation Agent — Solution Architecture

Component view and processing flow



The Reconciliation Service reads run results, exceptions and audit data from the Reconciliation DB.

01 Upload **02 Process (live progress feed)** **03 Review dashboard** **04 Resolve exceptions and export** **05 AI recommendation**

Matching rules: R Exact · RA Fuzzy / Amount · RE Reversal Auto-Pair · F Amount Difference · 1:M One-to-Many · M:1 Many-to-One

Exception categories: D · E · BAR · FR

Figure 2 — AI Reconciliation Agent: solution components and processing flow

Component	Function	Google Cloud implementation
Reconciliation Portal Front-end	Web interface for login, file upload, run execution, dashboard, exception queue and export	Firebase Hosting, with managed SSL and CDN
Reconciliation Service	Application and API layer; authentication, run management, persistence, exception presentation and report export	Cloud Run service
Reconciliation Orchestrator	Controls the reconciliation pipeline, sequences the processing stages and publishes progress	Module within the Cloud Run service
Data Cleansing Service	Parses the uploaded workbooks, normalises references, amounts and dates, and stages the data.	Module within the Cloud Run service
Rule-Based Reconciliation Engine	Executes the matching rules and produces matched pairs, groups and residual exceptions	Module within the Cloud Run service
AI Reconciliation Agent	Assesses each exception and generates an explanation and a recommended action	Module within the Cloud Run service
LLM and Reasoning Module	Large language model inference	Vertex AI — Gemini
Reconciliation DB	Stores staged ledger lines, match results, exceptions, explanations, users and audit trail	Cloud SQL for PostgreSQL
File store	Holds the uploaded Excel workbooks and generated reports	Cloud Storage

PROCESSING MODEL

Reconciliation executes within a long-running request on Cloud Run with CPU always allocated, allowing the Orchestrator to publish live progress to the front end as each stage completes — Ledger Parsed, References Normalised, Matches Found, Exceptions Categorized.

UPLOAD PATH

The front end requests a short-lived signed URL from the Reconciliation Service and uploads the workbooks directly to Cloud Storage. File contents do not pass through the application layer, which removes request size constraints and keeps application instances lightweight. The retained source files also allow any historical reconciliation run to be reproduced for audit purposes.

3.4 GOOGLE CLOUD LANDING ZONE

Abacus will provision a minimal landing zone sized specifically for this solution.

Region. All resources are deployed in me-central1 (Doha), the nearest Google Cloud region to the United Arab Emirates and a supported Vertex AI region.

Environments. Three Google Cloud projects are provisioned — Production, UAT and Development — each containing the application stack at the scale appropriate to its purpose.

Service	Purpose
Firestore Hosting	Front-end hosting, content delivery and managed SSL certificate
Cloud Run	Application, orchestration, cleansing, rule engine and AI agent
Artifact Registry	Container image repository
Cloud Storage	Uploaded ledger workbooks and generated reports
Cloud SQL for PostgreSQL	Reconciliation database
Vertex AI (Gemini)	Model inference for exception explanation and recommendation
Identity Platform	Portal user authentication and in-application user management
Cloud Identity	Administrator accounts for Google Cloud platform access
Secret Manager	Application credentials and configuration secrets
Cloud Logging and Monitoring	Operational logging, uptime checks and alerting
Security Command Centre	Baseline security posture visibility
Virtual Private Cloud	Standard regional VPC with a single subnet
Organisation policies	Resource location restriction, service account key restriction and external IP restriction

GOVERNANCE BASELINE

Resource creation is restricted to the nominated region. Service account key creation and external IP assignment on compute resources are disabled by policy. Access is granted through predefined IAM roles only, using dedicated service accounts for the application runtime and for deployment. Data at rest is encrypted using Google-managed encryption keys; data in transit is encrypted using TLS.

COST POSTURE

The Cloud SQL instance is provisioned as a single-zone instance with automated daily backups rather than a high-availability configuration. Cloud Run scales to zero outside active reconciliation runs. Vertex AI is consumed on a pay-per-use basis with no provisioned throughput commitment.

3.5 RECONCILIATION LOGIC

The reconciliation pipeline executes in the following sequence.

STAGE 1 — INTAKE

Both workbooks are uploaded and stored. The run is registered with a unique identifier, the reconciliation period start and end dates, and the identity of the submitting user.

STAGE 2 — PARSING AND CLEANSING

The Data Cleansing Service reads each workbook, locates the header row, maps columns to the canonical schema — reference, date, description, debit, credit and amount — and stages the rows. Cleansing operations include:

- Reference normalisation — case folding, removal of separators and whitespace, and correction of common character substitutions.
- Amount normalisation — a consistent debit and credit sign convention across both sides.
- Date parsing and standardisation.
- Removal of blank rows, subtotal rows and carried-forward balance lines.
- Duplicate detection within each ledger.

STAGE 3 — INTERNAL REVERSAL NETTING

Equal and opposite debit and credit entries within the AWS Distribution records are auto-paired and removed from the open item population before any comparison against the customer ledger takes place. This eliminates internal journal noise that would otherwise present as false exceptions.

STAGE 4 — RULE-BASED MATCHING

The engine applies the matching rules below in sequence. Once a transaction is matched it is locked and excluded from subsequent passes, so stricter rules always take precedence over looser ones.

Code	Rule	How it works
R	Exact Match	Reference matches after normalisation and the amount difference is under AED 1. Marked fully reconciled.
RA	Fuzzy / Amount Match	Identifies transactions that appear different across the two ledgers but are in fact the same transaction. Reference similarity of 80% or above by Levenshtein distance, combined with an amount match within a configured tolerance, allows the engine to recognise the same invoice or payment where the reference has been recorded inconsistently on either side — differing prefixes, separators, padding, character substitutions or truncation.
RE	Reversal Auto-Pair	Equal and opposite debit and credit entries within AWS Distribution records are auto-offset, removing internal journal noise before customer comparison.
F	Amount Difference	Reference matches but the amount differs beyond tolerance. Flagged for review — typically arising where the customer nets a deduction against the invoice.
1:M	One-to-Many	A single customer payment is identified as settling the sum of multiple open invoices.
M:1	Many-to-One	Multiple partial customer payments are grouped to settle a single invoice.

STAGE 5 — EXCEPTION CLASSIFICATION

Every transaction not matched by the rules above is classified into one of the following categories.

Code	Label	Trigger
D	AWS Distribution Only	Transaction present in AWS Distribution records but absent from the customer statement. Indicates a potentially unrecorded invoice or a payment missing on the customer side.
E	Customer Only	Transaction present in the customer ledger but absent from AWS Distribution records. May be a payment not yet posted, or a disputed deduction.
BAR	Posted After Cutoff	Entry date falls after the reconciliation period end date. A valid transaction but out of scope for the current period, carried forward.
FR	Rounding Difference	Amount difference under AED 1. Auto-flagged as rounding and does not prevent the reconciliation from being marked as balanced.

STAGE 6 — AI ASSESSMENT

For every flagged exception, the AI Reconciliation Agent submits the transaction, its counterpart candidates and the surrounding ledger context to the Gemini model through the LLM and Reasoning Module. The agent returns a plain-English explanation of why the item did not reconcile, a recommended next action for resolving it.

The AI layer does not alter the deterministic match results produced in Stage 4. It interprets and recommends; the matching arithmetic remains rule-based and reproducible.

STAGE 7 — PERSISTENCE AND REPORTING

All matches, groups, exceptions, explanations and recommendations are written to the Reconciliation DB against the run identifier. The dashboard presents the auto-match percentage, total matched value, and a breakdown of open exceptions by category. Users work the exception queue, approving or adjusting each item, and export the finalised reconciliation as an Excel report.

3.6 USER JOURNEY

Step	Stage	Description
01	Upload	User logs in and uploads the AWS Distribution Statement of Account and the Customer Ledger as Excel files.
02	Process	User executes the match engine and follows a live progress feed through parsing, normalisation, matching and exception categorisation.
03	Review	A summary dashboard presents auto-match percentage, total matched value and a breakdown of open exceptions by category.
04	Resolve and Export	User opens the exception queue, reads the AI explanation for each flagged item, approves or adjusts manually, then downloads the finalised Excel report.
05	Analyse	The AI agent assesses the remaining discrepancies and presents a recommended best next action for each.

3.7 SECURITY AND ACCESS MANAGEMENT

APPLICATION AUTHENTICATION

Portal users authenticate through Google Cloud Identity Platform, a managed Google Cloud identity service used here solely for application sign-in. Each environment holds its own private user pool, isolated from AWS Distribution's

corporate directory. Identity Platform manages password storage and hashing, password reset, session and token issuance, and protection against brute-force attempts. The application itself never stores or handles user passwords.

USER MANAGEMENT

Public self-registration is disabled; an account exists only where an administrator has created it. A small number of nominated users are granted administrator rights within the portal, allowing them to create users, issue set-password invitations, disable accounts and assign roles directly from the application interface. No access to the Google Cloud console is required to perform user administration, and no integration with an external identity provider or corporate directory is involved.

AUTHORISATION

Application roles and permissions — who may execute reconciliation runs, who may approve or adjust exceptions, and who holds view-only access — are held in the Reconciliation DB and enforced by the Reconciliation Service on every request.

PLATFORM ADMINISTRATION

Administration of the Google Cloud environment itself is performed separately, through Cloud Identity accounts under the AWS Distribution billing account, using predefined IAM roles and dedicated service accounts for the application runtime and for deployment.

DATA PROTECTION

- All traffic to the portal and application is served over TLS.
- Uploaded files are accessed exclusively through short-lived signed URLs; no object in the storage bucket is publicly readable.
- Uploaded workbooks are validated on the server for file type, size and structure before a run is accepted.
- Data at rest is encrypted using Google-managed encryption keys.
- All reconciliation runs, user approvals and manual adjustments are recorded in the audit trail against the acting user.

3.8 DELIVERABLES

1. Provisioned Google Cloud landing zone across Production, UAT and Development projects.
2. Reconciliation Portal front-end application.
3. Reconciliation Service, Orchestrator, Data Cleansing Service, Rule-Based Reconciliation Engine and AI Reconciliation Agent, deployed to Cloud Run.
4. Reconciliation database schema and deployment.
5. Vertex AI Gemini integration for exception explanation and recommendation.
6. Excel reconciliation report export.
7. Solution architecture and configuration documentation.
8. Administrator and end-user handover session, delivered remotely.

3.9 OUT OF SCOPE

The following are explicitly excluded from this engagement:

- Integration with FastERP or any other enterprise system, in either direction.
- Automated retrieval of ledger data from source systems; all files are uploaded manually by AWS Distribution users.

- OCR, PDF or scanned-document intake; Excel workbooks only.
- Integration with any external identity provider, corporate directory, single sign-on or federation; portal authentication is provided by Identity Platform only.
- Reconciliation scenarios beyond those defined in Section 3.5.
- A user-configurable rules engine allowing AWS Distribution to author or modify matching rules.
- Analytics, business intelligence or reporting beyond the dashboard and Excel export described above.
- Data migration or historical reconciliation backloading.
- On-site delivery, on-site training or on-site support.
- High availability, multi-region deployment or disaster recovery beyond automated database backups.
- Ongoing managed service, application support or cloud infrastructure operations after handover.

SECTION 04 – IMPLEMENTATION PLAN AND TIMELINE

4.1 DELIVERY APPROACH

Abacus will deliver the AI Reconciliation Agent over a six-week engagement, executed entirely off-site. Delivery follows an iterative approach: the reconciliation logic is built and validated against representative AWS Distribution data from the earliest possible stage, so that matching accuracy is proven progressively rather than discovered at user acceptance testing.

Provisioning of the Google Cloud landing zone runs in parallel with the opening development activity rather than as a sequential phase, allowing the full delivery window to be applied to the reconciliation engine and portal.

A structured internal quality assurance cycle is completed before the solution is released to AWS Distribution for user acceptance testing, ensuring that the acceptance window is spent validating business outcomes rather than identifying build defects.

4.2 IMPLEMENTATION PHASES

Week	Phase	Key activities
Week 1	Discovery and Solution Design	Requirements confirmation workshops; review of representative Statement of Account and Customer Ledger files; confirmation of matching rules, tolerances and exception categories; data model and solution design; agreement of acceptance criteria.
Week 2	Foundation and Build Commencement	Google Cloud landing zone provisioned across Production, UAT and Development projects; database schema deployed; application scaffolding, authentication and file upload established; data cleansing and parsing engine commenced.
Week 3	Reconciliation Engine	Data cleansing and normalisation engine completed; rule-based matching engine implemented — exact, fuzzy, reversal auto-pair and amount difference; grouping logic for one-to-many and many-to-one; exception classification.
Week 4	AI Agent and Portal	AI Reconciliation Agent and Vertex AI Gemini integration; exception explanation and recommendation generation; portal dashboard, live progress feed, exception queue and Excel report export; in-application user management.

Week	Phase	Key activities
Week 5	Testing and Quality Assurance	End-to-end testing; internal functional and accuracy testing against representative data; defect resolution; deployment to the UAT environment; preparation of test scenarios and user documentation.
Week 6	User Acceptance and Go-Live	AWS Distribution user acceptance testing; defect triage and resolution; production deployment; administrator and end-user handover session; solution documentation issued.

4.3 GOVERNANCE

- A weekly progress review is held with the nominated AWS Distribution business owner, covering completed activity, upcoming activity, risks and open decisions.
- A status report is issued weekly.
- Milestone deliverables are submitted for formal review and sign-off at the points identified
- Changes to agreed scope are raised through a change request and assessed for schedule and commercial impact before being adopted.

4.4 CRITICAL SUCCESS FACTORS

The six-week schedule is dependent on the following:

1. Representative Statement of Account and Customer Ledger files are provided at project kick-off, not during the development phase. Reconciliation accuracy is built and tuned against real data; late provision of sample files directly affects the delivery date.
2. Matching rules, tolerances, exception categories and the report format are confirmed and signed off by the end of Week 1.
3. A nominated AWS Distribution business owner is available throughout, with authority to make decisions and provide sign-off.
4. User acceptance testing resources are identified in advance and available for the full duration of Week 6.
5. Defects identified during user acceptance testing are triaged jointly. Items affecting reconciliation accuracy or preventing a reconciliation from completing are resolved before go-live; cosmetic and minor items are addressed under the warranty period.
6. Access and approvals required to enable Google Cloud services are provided within the first week.

SECTION 05 – GENERAL ASSUMPTIONS

1. All work is performed off-site. No on-site presence, travel is included.
2. The scope is as defined in Section 03. Any additional reconciliation scenarios, rules, integrations or features constitute a change request and will be estimated separately.
3. All Google Cloud resources are deployed in the me-central1 (Doha) or in agreed region during discovery phase. Should a Gemini model version required by the solution not be available in that region at the time of build, the Vertex AI global endpoint will be used and AWS Distribution will be advised in advance.
4. Three Google Cloud projects are provisioned — Production, UAT and Development. Non-production environments are provisioned at reduced capacity.

-
5. The Cloud SQL database is deployed as a single-zone instance with automated daily backups. High availability and cross-region disaster recovery are excluded. Final deployment metrics will be agreed upon during discovery and design phase
 6. Both ledger files are supplied as Excel workbooks in a consistent, agreed column structure. Material changes to the file format after the structure is agreed constitute a change request.
 7. File volumes are consistent with those observed during the Proof of Concept. Materially larger volumes may require resizing of the compute and database tiers, at additional Google Cloud cost.
 8. Reconciliation is performed for a single reconciliation period per run, between one AWS Distribution Statement of Account and one Customer Ledger.
 9. Matching tolerances, the fuzzy matching threshold and the reconciliation period cutoff are configured once during implementation to agreed values and are not user-editable thereafter.
 10. The AI agent produces explanations and recommendations only. It does not automatically post, adjust or approve reconciliation entries; all resolution decisions require user action.
 11. Outputs generated by the Gemini model are advisory. AWS Distribution retains responsibility for the accuracy and approval of the final reconciliation.
 12. AWS Distribution provides representative sample data for both ledgers at project commencement.
 13. AWS Distribution nominates a business owner empowered to confirm the reconciliation rules, exception categories and report format, and to sign off each delivery milestone.
 14. AWS Distribution nominates the administrator users to be provisioned at go-live.
 15. Portal users are managed within the application through Identity Platform and are separate from any corporate network or email credentials. Users will therefore hold a distinct set of credentials for the reconciliation portal. Single sign-on against a corporate directory is excluded.
 16. The solution is developed, tuned and validated against a maximum of **twenty (20) reconciliation scenarios (meaning AWS Ledger and customer ledger uploaded and recon taking place)**, agreed and documented in the scenario catalogue by the end of the Discovery and Solution Design phase. Additional scenarios constitute a change request and will be estimated separately.
 17. Uploaded workbooks to the portal should match the ones used during the PoC phase.
 18. AWS Distribution provides timely access to any approvals required to enable services.
 19. Review and sign-off of any deliverable is provided within five working days of submission. Delays in feedback may affect the delivery schedule.
 20. User acceptance testing is performed by AWS Distribution in the UAT environment against agreed test data and acceptance criteria.
 21. A warranty period covering defect resolution applies from go-live, on terms to be confirmed. Enhancements, new rules and new scenarios are excluded from warranty.
 22. Ongoing operational support and managed services for the solution and the Google Cloud environment are not included and can be proposed separately.